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S.340 CONCRETE WORK

1. SCOPE OF WORK

This section covers all concrete work including reinforcing steel, form, placing, finishing, curing, laboratory test, material test, providing ,batching plant, concrete mix design, mobilization, scaffold and other appurtenant work for buildings, structures and foundations as specified in Subsection 1.2 “Summary of Works”, including, but not limited to, the following:

- 1.1. Provision and installation of material for all expansion, contraction and construction joints.
- 1.2. Provision and installation of water stop.
- 1.3. Installation of anchor bolts and embedment’s in conjunction with concrete work.
- 1.4. For the purposes of this Technical Specification, the circulating water intake and discharge tunnel, circulating water pump houses and all structures in contact with sea water shall be defined as “the Marina Structures“.
- 1.5. Test of the concrete material: provide batching plant, concrete mix design and handling / storage of material.

2. CODES AND STANDARDS

Concrete Mix design and Handling / Storage of Material.

- 2.1. The Bidder shall comply with the requirements of Part 4, Subsection 2.5 “Codes and Standards”.
- 2.2. The codes and standards of the following organizations extracted from the Part 4, Subsection 2.5 “Codes and Standards ”are specifically applicable to the design, manufacture and testing of the work included in this Specification:

ASTM		American Society for Testing and Materials.
ACI		American Concrete Institute
CRSI		Concrete Reinforcing Steel Institute, U.S.A.

AASHTO		American Association of State Highway and Transportation Officials
USBR		U.S Bureau of Reclamation
BS		British Standard
PBI		Peraturan Beton Indonesia
SII		Standard Industri Indonesia
JASS		Japanese Architectural Standard Specific
JIS		Japanese Industrial Standards
AIJ		Architectural Institute of Japan
AWS		American Welding Standard

3. MATERIALS

Concrete shall be composed of cement, water, fine and coarse aggregate as specified herein under. The specifications for concrete are given in grades according to minimum strength tests at twenty eight (28) days, and maximum size of aggregates.

Materials used for the concrete work for this Project shall generally be product of the approved local factories.

3.1. CONCRETE

Material for the concrete work shall be approved by the Owner and shall conform to the requirements specified in Table S-340-1 or approved equivalent.

TABLE S-340-1. MATERIALS FOR CONCRETE WORK

ITEM NO	MATERIAL	ASTM OR BS STANDARDS	RESTRICTION OR EXCEPTIONS
1.	Cement	C 150	Type I for concrete work above ground level otherwise indicated. Type V for structures direct contact to sea water or ground water with similar behavior. Type II for substructures.
2.	Fine Aggregate	C33, C289, C295, BS 882	Use only graded natural sand unless otherwise indicated.
3.	Coarse Aggregate	C33, C289, C295 BS 882	Use only graded crushed stone unless otherwise indicated with sized as specified in Table S-340-4.
4.	Reinforcing Bars (deformed bars)	4.1 A615 Grade 60 4.2 A615 Grade 40	For all reinforcing unless otherwise indicated. For dowels of equipment foundations over grade slab only where specifically indicated.
5.	Polypropylene fibre	C39, C78, C496	For all important structures' concrete if required, see Clause 3.2.
6.	Bar Support		CRSI manual of Standards Practice Class C. For mat foundations on soil or mud mats, use precast concrete blocks having the same compressive strength as the mat foundation.
7.	Form Ties		Steel snap-ties architectural type without washers, or better.
8.	Welded Steel Wire Fabric	A185 ; AWS 012.1	Spacing and gauge of wires as indicated on the Bidder's design drawings.
9.	Air-Entraining Admixtures (AEA)	C260, BS5075	Total air percent by volume of concrete shall be to Owner's approval.
10.	Water-Reducing Admixtures (WRA)	C494, BS5075	Set retarding agent or approved equivalent to pozolith 300R liquid. Type D shall be used when the weather is such that the temperature of concrete would exceed 30°.
11.	Silica Fume Mineral Admixture (IMB-SF)		To be used per manufacturer's instructions in all foundations, and all other work, (wax base not permitted).
12.	Membrane Curing Compounds	C309	Type 1, for concrete floor and tread finish , type 1-D with fugitive dye for all other work, (wax base not permitted).
13.	Water	C94	To be fresh, clean and drinkable.
14. 14.1	Forms Plywood (exposed surface)		Waterproof, resin bonded, exterior type
14.2	Lumber		Straight, uniform width and thickness, and free of knots, offsets, holes, and other surface defects
14.3	Chamfer Strips (beveled corners)		Wood, surface against concrete, planed
14.4	Metal Deck		(See Specification S-406)

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3.2. POLYPROPYLENE FIBRE

Polypropylene fibres as the material may be used, if the Bidder intends to use.

Polypropylene fibres shall be 100% polymerized virgin polypropylene collated which are chemically inert, unaffected by water, alkalis in cement or chemical admixtures, have no effect on the chemistry of the concrete mix, and have no abnormal effect on the finished quality of concrete during curing.

This fibre shall be mixed in the concrete for the structures, If required

Mix proportion of this material shall be based on the manufacturer's recommendation and on trial tests of concrete strength.

Polypropylene fibres selected shall be subject to the Owner's approval and shall have minimum physical properties such as zero water absorption, specific gravity of 0.9, melting point at 160°C, ignition point at 590°C, tensile strength of 560 MPa, Young's modulus of elasticity of 3500 MPa, and high resistance to acid and salt.

3.3. ADMIXTURES

The use of admixtures shall be approved by the Owner. Admixtures containing calcium chloride in any form shall not be used for the concrete work at or below grade or concrete in contact with sea water. All admixtures shall be compatible with all other admixtures used.

All admixtures shall be added to the batch so as to ensure uniform distribution throughout the batch.

Manufacturer's certificates of compliance with this Specification shall be supplied by the Bidder for each admixture.

The Bidder shall on the direction of the Owner perform tests on admixtures both before and after delivery to demonstrate compliance with ASTM Specifications C260 and C494 or JIS or approved equivalent as applicable.

3.4. PRELIMINARY REVIEW

The source and quality of concrete materials and the concrete proportions proposed for the work shall be submitted to the Owner for review and approval before the concrete work starts. Complete certified reports prepared by an independent testing laboratory and covering the materials and proportions shall be submitted to the Owner for approval.

3.5. REPORTS

- 3.5.1** Mill Test Reports of Reinforcing Bars: These reports from the approved manufacturer shall accompany each delivery of reinforcing bars, indicating that these bars have been tested in accordance with ASTM A615. At the same time, two copies of these reports shall be submitted to the Owner for information.
- 3.5.2** Reports from the approved cement manufacturer shall accompany each delivery of cement and shall indicate the type, brand, manufacturer, composition and method of handling (scale or bulk) and compliance with ASTM C150 and all other requirements of this Specification.
- 3.5.3** Reports on fine aggregates provided shall include the source, type, gradation, deleterious substances, soundness, abrasiveness, absorption rate and the results of all tests required to verify compliance with standards specified.
- 3.5.4** Reports on coarse aggregates shall include the source, type, gradation, deleterious substances, soundness, abrasion loss, absorption rate and the results of all test required to verify compliance with standards specified.
- 3.5.5** Reports on admixtures shall include the applicable ASTM C260 or C494 classification, brand, and manufacturer, character of the material and lot number. All admixtures shall be the product of one manufacturer.
- 3.5.6** When required by the Owner, the Bidder shall at his expense have the water test in accordance with ASTM C94, and the reports shall be submitted to the Owner.

4. STORAGE OF MATERIALS

- 4.1. Store cement at all times in a manner approved by the Owner. The cement storage area for bagged cement shall be a weather proof building, well ventilated, lighted and free from dampness. The size of storage shall be sufficient for 2 weeks' continuous output of the batching plant at its designated capacity. Cement delivered in bulk shall be stored in light colored silos or sheds built for this purpose and approved by the Owner.

Any bag, package or container that has been damaged or any consignment of cement that shows sign of deterioration shall be rejected and the Bidder shall remove all rejected cement from the site without delay.

- 4.2. The cement shall be used in the order of its delivery and shall be stored such that each delivery is readily distinguishable from previous deliveries.
- 4.3. The Bidder shall keep a record of the date, quantity, storage location and place of use in the Plant for each delivery of cement. The Bidder shall also keep a record showing the stock of cement held at all times.
- 4.4. Store aggregates in a manner that will preclude deterioration or inclusion of foreign material and contact with sea water. Store aggregates of different sizes in separate piles in a manner to avoid segregation. If segregation does occur, remix aggregate to conform to the specified grading requirements.
- 4.5. Store reinforcement in a manner that will avoid excessive rusting, coating with grease, oil, dirt and/or other objectionable materials. All reinforcement shall be stored on the Site on timber or concrete supports suitably spaced and of sufficient height to keep the steel clear of the ground.
- 4.6. Store admixtures in such a manner as to avoid contamination, evaporation or damage.

5. CONCRETE STRENGTH AND PROPORTIONS

5.1. GRADES OF CONCRETE

The grades of concrete shall comply with the following table having the minimum compressive strength shown.

MINIMUM		
GRADE	COMPRESSIVE STRENGTH	USE
15	15N/mm ² (150 kg/cm ²)	Lean Concrete
25	25N/mm ² (250kg/cm ²)	General Structure
30	30N/mm ² (300kg/cm ²)	Important structures* And Marine Structures (see Clause 1.4)
50	500 kg/cm ² (500kg/cm ²)	Pre-cast Concrete Pile

* Steam turbine foundations, steam turbine building, central control building.

The compressive strength shall be determined by ASTM C 39 at 28 days.

5.2. LIMITING REQUIREMENTS

Each concrete mix shall be designed and concrete shall be controlled within the following limiting requirements:

- 5.2.1** The acceptability of concrete shall be judged on the basis of compliance with all of the specified requirements and not on the basis of strength alone.
- 5.2.2** Coarse aggregate sizes described hereinafter are the nominal sizes given in Table S-340-4. To qualify as a certain aggregate size, the gradation of the aggregate must be within the tabulated limits in ASTM C33.
- 5.2.3** The potential alkali reactivity of the aggregates and the concrete mixes shall be determined by the tests in accordance with the requirements of ASTM C227, C289 and C295.

5.2.3.1 Chlorides

The amount of the total water soluble chloride ion content of any mix shall not exceed the following:

- a. Reinforced concrete : 0.15 % by weight of cement
- b. Mass concrete without reinforcing : 1.00 % by weight of cement
- c. Prestressed concrete : 0.06 % by weight of cement
- d. The chloride ion content in cement, mineral admixtures, aggregates (test method ASTM D1411), water (test method ASTM D 512), air entraining admixtures and chemical admixtures shall be included in calculating the total water soluble chloride content.

5.2.3.2 Sulphates

- a. If sulphates in the aggregate are detected, the aggregates shall be washed to eliminate the water soluble sulphate content.
- b. The total sulphate content of any mix excluding that present in the cement but including any present in other ingredients shall not exceed 0.5 by weight of cement in the mix.

5.2.4 The 38 mm size coarse aggregate shall not be used in concrete which has clear reinforcing spacing less than 50 mm.

5.2.5 Minimum Cement Factors

The quality of Portland Cement, expressed in kg per cubic meter, shall not be less than that indicated in the table as shown in S-340-3.

A sack of cement in Indonesia is 40 kg.

5.2.5.1 Multilayer flooring

Multilayer flooring shall be applied in the battery room floor area as indicated in the Finishing Schedule attached in S001 or as shown on the Bidder's drawings.

The epoxy should have chemical resistance to alkalis, acids, solvents, beverages with the material density and bonding strength of not less than 1.3kg/cm³ and 25 kg/cm² respectively.

5.2.5.2 Application

Prior to use of multilayer flooring, the surface shall be cleaned, dried, sound and free from contaminated materials. Large cracks and holes shall be filled with an approved material. The placing procedures shall be in accordance with the manufacture's instruction.

TABLE S-340-3

MINIMUM & MAXIMUM CEMENT CONTENT W/C RATIO

CONCRETE GRADE	MINIMUM CEMENT CONTENT IN Kg/m ³	MAXIMUM WATER/CEMENT RATIO BY WEIGHT
15	200	0.52
25	275	0.50
30	300	0.50 (Note 1)

Note : 1. The water/cement ratio shall not exceed 0.40 by weight for concrete used in marine structures (see Clause 1.4)

5.2.6 Water

The water shall be fresh, clean and potable. The water chemistry shall be in accordance with the following:

- 5.2.6.1 Total solid per AASHTO T.26
- 5.2.6.2 Chlorides per ASTM D512
- 5.2.6.3 Sulphate less than 500 mg/liter
- 5.2.6.4 Compressive strength of 50 mm cubes made with water must be within 10% of the strength of 50 % of the strength of 50 mm cubes made using distilled water.

5.2.7 Fine and Coarse Aggregate

In accordance with ASTM C33, Tables S-340-1, S-340-4 and the following:

- 5.2.7.1 Elongation index to be less than 30%.
- 5.2.7.2 Flakiness index to be less than 25%
- 5.2.7.3 Water absorption to ASTM C127 to be less than 2%
- 5.2.7.4 The severe weathering regions in Table 3 of ASTM C33.
- 5.2.7.5 Magnesium sulphate soundness loss to ASTM C688 to be less than 14% (compared to 15% and 18% limit of ASTM C33).
- 5.2.7.6 The apparent specific gravity of aggregate as determined by an approved test, such as in ASTM C127, shall not be less than 2.5.
- 5.2.7.7 10 % fines value to BS 812 to be less than 8 t.

TABLE S-340-4 GRADING REQUIREMENT

ITEM NO	GRADING REQUIREMENT	GRADING REQUIREMENTS – ASTM C33	
		Nominal Size (Sieves)	Size Number
1.	All concrete 300 mm or less in thickness, unless otherwise indicated	19 mm to No. 4	64
2.	All other concrete work, unless otherwise indicated	38 mm to No.4	467

5.2.8 Proportions and Slump

- 5.2.8.1 The proportions of cement to coarse aggregate shall be the least which will produce a plastic mix of suitable workability for each portion of the work so as to result in uniformly dense concrete free from aggregate pocket or honeycomb.
- 5.2.8.2 Slump shall be 80 mm to 125 mm
- 5.2.8.3 Proportions to meet these requirements shall be established by design mixes in accordance with the Chapter 4 of ACI 318 and all concrete for the work shall be in strict conformance with these design mixes.

5.2.9 Air Entrainment

- 5.2.9.1 An air-entraining admixture conforming to the requirements of Sub clause 3.3 admixtures or approved equal to liquid AEA-303 shall be added at the mixer in such quantities as to produce concrete with total air content conforming to the moderate exposure requirements of Table 4.5.1 of ACI 18-83. The admixture shall be added as a solution in a portion of the mixing water, using a suitable mechanically activated dispenser.
- 5.2.9.2 At the Site as directed by the Owner, field tests for air entrainment shall be carried out by the Bidder at his own cost in accordance with ASTM Specification C-260 or BS.5075.

5.3. CONCRETE DESIGN MIXES

- 5.3.1 The Bidder shall furnish to the independent_testing laboratory sufficient quantities of cement, aggregates, water and admixtures, polypropylene fibre, from the same sources as will be used for the concrete work on Site, for testing of these materials and preparation of trial mixtures. At least 20 pieces of sample cylinder for each concrete grade shall be prepared for mix proportion according to Indonesian Concrete Standard (PBI). The materials thus furnished shall be used for the entire Project without changes. Changes in the source of the materials shall require the preparation of trial mixes and a revised mix design, including laboratory test for material, which shall be approved by the Owner.
- 5.3.2 Trial mixtures shall be designed to produce slumps of 80 mm to 125 mm and within 0.5 percent of the maximum allowable air content. For each proposed mixture at least three compressive test cylinders for each age shall be made and cured in accordance with ASTM C192. The cylinder shall be tested for strength in accordance to ASTM C39 at 3, 7 and 28 days. From the test result curves shall be plotted showing the water-cement ratio relationship with the strength. Using these curves concrete proportions shall be selected to produce the average strength required by ACI 318.
- 5.3.3 The Bidder shall submit the tests result of design mixes to the Owner at least two (2) weeks before starting the concrete work.

6. FORM WORK

6.1. GENERAL REQUIREMENTS

Forms shall be designed to produce hardened concrete having the shape, lines, grade and dimensions indicated on the approved Bidder's drawings and conform to the applicable requirements of ACI 301 and to the followings:

6.1.1 Build form true to the line and grade of matched surface of lumber, plywood, steel or other materials. The forms shall be properly supported, braced and tied to maintain position, shape, and alignment during and after placing concrete therein.

6.1.2 Wales, studs, internal ties, and other form supports shall be of sufficient size and number and shall be located and spaced so that proper working stresses therein are not exceeded.

Temporary shoring for suspended elements shall not be disturbed during removal of column forms.

6.1.3 The top edges of the forms for walls, equipment bases and other concrete which is to be finished to a specified elevation, slope or contour, shall be brought to a true line and grade so that the top concrete surface can be finished with a screed or template resting on the top edges of the forms. Wall forms, on one side at least, shall not extend above horizontal construction joints.

6.1.4 Completed work shall have a smooth finish and uniform color, and shall be level, plumb and true. Forms shall be substantial and sufficiently tight to prevent leakage of mortar. When forms for slabs and roofs in the form of metal decking are used, the Bidder shall provide suitable closures to prevent leakage of mortar.

6.1.5 Arrange bolts and rods used for internal ties so that all metal remaining in concrete shall not be less than 25 mm from concrete surface after ties are removed for onshore work and 75 mm for marine work. Twisted wire ties shall not be used

6.1.6 Provide temporary openings at base of forms for columns, walls, piers, and at other points where necessary to facilitate cleaning and inspection prior to placing of concrete.

6.1.7 Cleaning and Treatment of Forms

No concrete shall be placed in any portion of the work until the forms for the position have been completed and reviewed by the Owner. All rubbish shall be removed from the interior of forms before the concrete is placed.

The faces of the form in contact with the concrete shall be clean and treated with a suitable release agent. Forms that are to be clean and concrete surfaces on which tile adhesives, paint, epoxies, masonry coatings or overlays are to be applied, shall not be treated with form coating or release agents that leave residues on the surfaces on the concrete. The release agent shall be chosen by the Bidder and approved by the Owner. Only one agent shall be used for all surfaces that are permanently exposed.

6.1.8 Removal of Forms

Do not remove forms work of any concrete work until the concrete has acquired sufficient strength to safely carry its own weight and any construction loads that may be imposed on it. Methods used for removal of form work shall be such as to prevent marring, breakage or other damage to the concrete. Removal of form shall conform to the applicable requirements of ACI 301 or JIS or approved equivalent.

The formwork shall not be removed for at least the periods shown in the following table.

TABLE S-340 – 5 MINIMUM FORM WORK REMOVING

TYPE OF FORMWORK	MINIMUM PERIOD BEFORE REMOVING
Vertical formwork to columns Walls and large beams.	2 days
Soffit formwork to slabs.	4 days
Soffit formwork to beams.	8 days
Formwork to slabs.	11 days
Formwork to beams.	15 days

Formwork for any concrete which will be immersed in sea water shall not be slackened or removed in less than 7 days (or for the longer duration per above).

The formwork to vertical surfaces such as beam sides, walls and columns shall not be removed in less than 2 days without Owner's written approval. When doing so, extra care shall be exercised to avoid damage to the concrete.

6.1.9 Submittals

Proposals for the formwork of the structures shall be submitted to the Owner for review.

6.2. FORMS FOR EXPOSED SURFACES

6.2.1 "Exposed Surfaces" shall mean all formed concrete surfaces exposed to view on completion of work, unless otherwise indicated.

6.2.2 Forms for all exposed surfaces shall produce extremely smooth, dense and true finished free of fins, imperfections or other defects. For exposed surfaces of structures in contact with ground, such finish shall extend 300 mm below finish grade

6.2.3 Forms shall consist of plywood, lined forms or accepted equivalent. Plywood shall be water resistant and shall be treated to prevent rising of the grain. Lining shall consist of accepted non absorptive fibre board with a plastic coated hard smooth surface on the contact side to resist moisture.

6.3. BEVELED CORNERS

Bevels on all exposed projecting corners of concrete work, such as piers, columns, equipment foundations, etc. shall be 25mm by 25mm unless otherwise indicated. For exposed vertical corners in contact with ground, extend bevels 300mm below finish grade.

6.4. COATING OF FORMS

Coat forms for exposed concrete surfaces with non-staining mineral oil, applied before reinforcement is placed. For unexposed concrete surfaces, forms may be wetted thoroughly with water.

6.5. CONCRETE TOLERANCE

For concrete works, the departure from alignment, grade and level shall not exceed the following or the regulation specified ACI 117

- Level of concrete or reinforced concrete structures : ± 20 mm
- Cross section dimensions of structures : ± 10 mm
- Variation from the plumb in columns, piers and walls : in 3 meters 6 mm more than 25 mm
- Level of sills, orifices, embedded pipes etc : ± 10 mm

The values are general tolerances for concrete. The tolerance for foundations or support of special equipment shall be fixed by the Bidder and applied whenever necessary.

7. CONSTRUCTION DETAILS

7.1. OPENINGS IN CONCRETE

- 7.1.1** Provide openings, pockets, chases, recesses, slots etc. in walls, piers, floors or other portions of the concrete work, conduit, pipe, sewers, sleeves and similar items, as may be required.
- 7.1.2** Fill all such openings, pockets, chases etc. with concrete after preceding item, are installed as the work progresses. Finish exposed surfaces of such concrete fill to match adjoining concrete work
- 7.1.3** Concrete wall and floor openings for piping and other fixtures which will be installed after the walls and floors are built shall be of sufficient size that adequate space is available for the proper compaction of concrete which will be placed later around the pipe or fixture. All sides of each opening shall be provided with continuous keyways. The top of each wall opening shall be sloped or beveled to provide adequate space for placing and compacting the pipe embedment concrete.

7.2. EMBEDDED MATERIALS

Embedded materials shall include all work, which is required by the Bidder, and also all pipe, conduit, ground bars, electrical equipment, etc. Use care in placing concrete so as not to dislodge any of these installation.

7.3. CONSTRUCTION JOINTS

7.3.1 All construction joints as required by the Bidder shall be shown on the Bidder's design drawings. Construction joints shall be prepared by brush hammering, providing a rough, coarse surface with aggregate exposed. All loose material shall be removed by blasting and the surfaces to be joined shall be presoaked or coated with an approved bonding agent prior to placement of new concrete.

7.3.2 If the Bidder desires to revise locations of constructions joints or to add or delete construction joints, the drawings showing proposed changes shall be submitted to the Owner for approval. All changes, including required water stops, if any, shall be made without additional cost.

7.3.3 Waterstops shall be clean and free from coating that would weaken the bond with concrete. Each waterstop shall be continuous throughout the length of the construction joint in which it is installed. Junctions between adjacent section shall be spliced in strict conformity with the recommendations of the waterstop manufacturer. All waterstops shall be maintained in proper position until the surrounding concrete has been deposited and compacted.

7.4. EXPANSION JOINTS

The Bidder shall submit drawings showing the location of all expansion joints to the Owner for approval. Expansion joints shall be located and constructed as indicated on the Bidder's drawings. Expansion joint filler (premolded type) shall be firmly bonded to the previously poured joint face with a suitable adhesive and the new concrete shall be poured directly against the joint filler.

Accessible edges of each expansion joint shall be sealed with rubber sealer applied in accordance with the manufacturer's directions.

Expansion joints so designated on the Bidder's drawings, and which will be subject to the differential hydrostatic pressure, shall be provided with a flexible waterstop. Waterstop embedment shall be equal on each side of the joint.

Flexible waterstops shall be spliced in strict conformity with the recommendations of the approved waterstop manufacturer.

Rubber and plastic materials shall be stored in a cool place and shall not be exposed to direct sunlight.

7.5. CONNECTIONS TO EXISTING CONCRETE

Connections of new concrete to the existing concrete shall include all necessary cutting, patching and fitting required for a neat and workmanlike job. All exposed surfaces shall be cut true and smooth and shall be patched and finished to match adjoining concrete surfaces.

7.6. RUBBER WATERSTOPS

7.6.1 General

Flexible waterstops of rubber type shall be placed across the joints in concrete below grade level. The waterstops shall be suitable for the hydrostatic pressure and for sea water environment.

7.6.2 Material

The rubber waterstops shall be made from a high-grade, tread-type compound. The basis polymer shall be natural rubber or synthetic rubber conforming to the requirements of CRD-C513. The Bidder shall provide standard factory fabricated T shape, L shape, X shape and other joint fittings required for all intersections. The waterstop must show a tensile strength of more than 130 kg/cm² and an elongation before rupture not less than 320 %.

7.6.3 Type and Make

The waterstops shall be 250 mm minimum wide, center bulb flat type for expansion joints, and flat dumbbell type for all other joints.

7.6.4 Fabrications

The waterstops may be molded or extruded. All materials shall be molded or extruded and cured in such a manner that any cross section will be dense, homogeneous and free from porosity and other imperfections.

7.6.5 Tests and Test Report

The waterstops shall be subject to the laboratory tests prior to delivery to the Project Site. Testing of waterstops shall conform to the requirements of CDR-C513. The waterstops shall be capable of withstanding a maximum design hydrostatic head of 20 meters. The approved manufacturer shall furnish certified copies of all laboratory tests reports.

7.6.6 Installation

7.6.6.1 The waterstop shall be installed with approximately one-half of its width embedded in the concrete on each side of the joint.

Nailing of waterstop to formwork shall not be permitted. Care shall be exercised in placing and vibrating the concrete about the waterstop, to insure complete filling of the concrete around the waterstop, and to obtain a continuous bond between the concrete and the waterstop at all points around the periphery of the waterstop. Rubber unions and rubber cement shall be used for splicing these waterstop at all joints and at fittings splicing shall be done in accordance with the manufacturer's instructions.

7.6.6.2 The Bidder shall take suitable precautions to support and protect the waterstops during the progress of the work, and shall repair or replace any damaged waterstop as directed by the Owner. All waterstops shall be protected from oil or grease.

7.6.6.3 In the event the waterstop is installed in the concrete on one side of a joint more than one month prior to the scheduled date of placing concrete on the other side of the joint, the exposed waterstop shall be covered or shaded to protect it from the direct rays of the sun during exposure.

7.6.6.4 The Bidder shall provide the services, without additional cost to the Owner, of a technically qualified manufacturer's field representative to instruct Bidder's field installation personnel on the proper procedures for the installation of the waterstop.

8. FABRICATION AND PLACING OF REINFORCEMENT

8.1. MATERIALS INCLUDED

Reinforcement shall include all reinforcing bars, stirrups, spirals, and other reinforcement materials with all necessary wire ties, bar supports, spacers, block supports and other devices required to install and secure reinforcement properly.

For the reinforcement bars, plain bars shall be dia 6, 8 and 10 mm and deformed bars shall be D10, 13, 16, 19, 22, 25, 29 and 32 mm unless otherwise required.

8.2. FABRICATION

8.2.1 The details of fabrication shall conform to ACI 315 and 318. in case of conflict, ACI 318 shall govern.

8.2.2 Cutting and Bending

Cutting, bending and tolerances shall be in accordance with CRSI Manual of Standards Practice.

Reinforcement may be supplied pre-bent from the approved manufacturer or in stock lengths and cold bent on Site. Bending of reinforcing bars in hardened concrete will not be permitted.

8.2.3 Heating and Welding

Heating and welding of reinforcement for any purpose, and tack welding in particular, is expressly prohibited.

Bar support may be used provided that they are clamped or wired to the reinforcement.

8.2.4 Splicing of Reinforcing Bars

Shall conform to the applicable requirement of ACI 318, unless otherwise indicated. Mechanical splices shall be in accordance with the manufacturers' recommendations.

Splices shall not be used in regions of maximum bending stress.

8.3. CONCRETE COVER

Thickness of concrete cover shall be in accordance with the applicable requirement of ACI 318 or JASS 5 or CRSI Manual of Standard Practice, unless otherwise indicated. Concrete cover shall be clearly indicated on the Bidder's drawings.

Except as otherwise indicated on the Bidder's drawings, metal reinforcement for concrete shall have the following minimum covering depth:

TABLE 340-5 MINIMUM CONCRETE COVERING DEPTH

No	Type of Concrete Placing	Minimum Concrete Covering Depth
1	Contact with the ground surface	Not less than 70 mm
2	Marine Structure	Not less than 70 mm
3	Exposed to the weather and others	30 mm or 2.5 D (D is max. diameter/out side re-bar)
4	Below grade level	75 mm
5	Not exposed to the ground or weather	Not less than 25 mm

8.4. PLACEMENT

All reinforcement shall be free of loose rust or scale, grease, dirt, or any other coating that would reduce or destroy the bond. Reinforcement shall be accurately positioned on supports, spacers, hangers, concrete blocks or other reinforcement, and shall be secured in place during the course of construction.

With the exception of lapped portions of spliced bars which are wired or clamped together, the clear distance between parallel bars shall be not less than 1.5 times the maximum size of coarse aggregate in the concrete, or more than 25 mm in beams, 40 mm in columns, or 50 mm in other locations. Where reinforcement in beams is placed directly above the bars in the lower layer and the clear distance between bars shall be not less than 25 mm.

8.5. PLACING OF WELDED WIRE FABRIC

Carefully place fabric reinforcement and maintain in a position before and during placing of concrete.

8.6. ANCHORING DOWELS

Securely anchor all dowels in place by wire tying, etc., before starting placement of concrete.

8.7. COATING DOWELS FOR FUTURE EXTENSIONS

Exposed reinforcing bars and metal waterstops intended for bonding with future work shall be protected from corrosion with two coats of bitumastic. Before coating the dowels, apply two coats of bitumastic concrete penetrant on the surface adjacent to the dowels. When the dowels are coated, the coated concrete surface shall be included as an integral part of dowel coating.

9. FIELD CONTROL TESTING

9.1. GENERAL REQUIREMENTS

9.1.1 Field control tests shall cover all material and concrete test required, but not be limited, to the following:

9.1.1.1 Slump tests (ASTM C143), every mixer truck, before concrete pouring (every 5 m³ or less)

- 9.1.1.2 Air content tests (ASTM C231), every 50 m³
- 9.1.1.3 Aggregate gradation tests (ASTM C136), daily
- 9.1.1.4 Specific gravity tests (ASTM C 127), weekly
- 9.1.1.5 Organic impurities (ASTM C40), daily
- 9.1.1.6 Surface moisture (ASTM C 70), daily
- 9.1.1.7 Concrete compressive strength test (ASTM C 39), every 50 m³ or less
- 9.1.1.8 Concrete temperature, every 50 m³
- 9.1.2** The Bidder shall appoint at his own expense an approved independent material and concrete testing laboratory fully equipped to carry out all testing required to meet the requirement of the ASTM and ACI.

The laboratory must provide evidence that his equipment and procedures are complied with ACI or other approved standards and that its equipment is regularly checked, calibrated and adjusted for accuracy. If requested, full details of the professional laboratory and qualifications of all laboratory staff shall be submitted to the Owner.
- 9.1.3** The Bidder shall provide the services of one or more qualified employees as required for marking the test and test specimens and for storing, curing, handling and delivery of tests specimens to the independent laboratory.

All field control tests and the preparation of concrete test specimens shall be made by the Bidder's employees in the presence of the Owner.
- 9.1.4** All frequencies specified for each field control test are minimum. If any additional field control tests are necessary, in the opinion of the Owner, the Bidder shall carry out the additional tests without additional cost to the Owner.
- 9.1.5** All the tests reports shall be sent directly to the Owner from testing laboratory within 48 hours of the tests. The Owner on Site shall have the authority to stop all further concrete work until acceptable tests results are forth coming. Up-to-date record shall be kept by the Bidder of placement locations of every batch of concrete, of its grade and of all test cylinders, core, or other specimens taken. Copies of these records shall be supplied to the Owner at weekly intervals or upon request by the Owner.

9.2. AGGREGATE GRADATION

Fine and coarse aggregate shall be sampled and tested in accordance with ASTM D75 and C136.

9.3. SLUMP TEST

Slump test (ASTM C143) shall be made at every truck mixer and less than 5 m³ and shall be made from the similar batch mixed each day prior to concrete placing.

Each set of compression test cylinder shall be marked or tagged with the slump.

9.4. AIR CONTENT

An air content test shall be made from the first batch mixed each day and from each batch of concrete from which concrete compression test cylinders are made. Air content shall be determined in accordance with ASTM C231.

Each set of compression test cylinders shall be marked or tagged with the air content.

9.5. CONCRETE TESTS

9.5.1 Production Test

- 1) To verify the compatibility of the laboratory design mixes with the working conditions on Site, the Bidder shall carry out trial mixing under full scale production conditions.
- 2) Procedures and number of test cylinders to be tested shall be as for the laboratory design mix.
- 3) Trial mixing under full scale production conditions shall be carried out at least two (2) weeks before commencing any concreting at the site, under supervision of the Owner.

9.5.2 Concrete Testing

- 1) Cylinder moulds, curing and test specimens shall be in accordance with ASTM C31, C39, C143, C172, C231 and C511. The Bidder shall provide cure boxes for storing test specimens, meeting the temperature and other requirement of ASTM C31.

- 2) The testing laboratory will provide cylinder moulds and instructions to the Bidder for properly making and storing job concrete test cylinders, instructions for checking percentage of air entrainment, and for making slump tests.

9.5.3 Job Control of Concrete

- 1) Control of concrete mixes for the site work shall be performed by the Bidder as follows, subject to review by the Owner.
- 2) The Bidder shall make all job concrete test cylinder, as specified in Table S-340-5, from fresh samples of concrete taken on site from discharge of stationary mixers, Concrete pumping machines, truck mixers or truck agitators. Locations for taking samples shall be acceptable to the Owner.
- 3) Making of test cylinders shall include slump test, staring, packing, delivery to testing laboratory of test cylinders and all required data as to materials, water-cement ratio, proportions, slumps etc.

TABLE S-340-6 SCHEDULE OF JOB CONCRETE TEST CYLINDERS

Total cylinders required from a sample representing each 50 cu m of concrete, or from each day's pour if less than 50 cu m.	Cylinders for strength test at (for grade 15, 25 & 30)		
	3 days	7 days	28 days
3 sets of 3 Each – total, 9 each concrete grade	3	3	3

NOTE: One set of 3 cylinders comprises a strength test.

- 4) Unless otherwise indicated, the Bidder shall at his own expense furnish equipment for and shall check the slump of concrete and air content of air-entrained concrete by direct measurement of fresh samples of concrete taken on site from discharge of stationary mixers, concrete pump machines, truck mixers or truck agitators.
- 5) Equipment shall consist of slump cones, one air meter and one or more pocket type air indicators. Locations of samples for testing slump and air entrainment shall be the same as for samples for test cylinders.

- 6) Making of air tests shall include delivery to testing laboratory of all required data as to materials, name and quantity of air entraining admixtures, water-cement ratio, proportions, slumps, air content, equipment with which tested.

10. COMPLIANCE WITH STRENGTH REQUIREMENTS

10.1. COMPRESSIVE STRENGTH

Compliance with the specified compressive strength shall be judged by tests made on cylinders at an age 28 days.

10.2. STRENGTH TEST RESULTS

Shall conform to the requirement of Table S-340-7.

TABLE S-340-7 REQUIREMENT FOR STRENGTH TEST RESULTS

ITEM NO	TEST RESULTS	REQUIREMENTS
1.	Average of any 3 consecutive strength results	Equal to or greater than specified strength
2.	For any strength test result	To be subject to the Owner's direction

10.3. CONCRETE BELOW SPECIFIED STRENGTH

- 1) If the results of strength test indicate that concrete does not meet specified requirements the concrete production shall be stopped until corrective action is implemented to guarantee conformance with this Specification.
- 2) The Bidder may be directed to take concrete cores and/or make load tests, at his own expense, of any work that he has installed with concrete not meeting specified strength requirements.
- 3) If cores and/or load tests establish that such work in place does not meet requirements, the Bidder shall correct the work as requested without cost to the Owner.

11. MIXING, CONVEYING AND PLACING CONCRETE

11.1. MIXING AND EQUIPMENT

11.1.1 Equipment

- 1) Batching and mixing equipment shall be approved to the Owner. Equipment shall be to the site sufficiently in advance of its use to permit inspection and acceptance. Ready mixed concrete factory can be used as an alternative.
- 2) The concrete shall be mixed in pan or drum type mixers and the weight of materials used for each batch shall not exceed the manufacturer's rated capacities for the machines. Other types of mixer shall be used only after permission has been obtained from the Owner.
- 3) All mixing machines shall be kept clean, free from hard or partially set concrete and well maintained in good working order.
- 4) Mixers and agitators shall be examined from time to time for changes in condition due to the accumulation of hardened concrete or mortar or through wear of the blades, and if found defective shall be corrected by the Bidder at Bidder's cost.
- 5) The materials used in concrete shall be measured solely by weigh, by means of an approved weight batching machine. Due allowance shall be made for the water content of the aggregates.

11.1.2 Mixing

- 1) All concrete shall be mixed until there is a uniform distribution of materials and shall be discharged completely before mixer is recharged. Production of concrete shall be in accordance with the applicable requirements of ACI 301 and ASTM C94.
- 2) The time allowed for mixing after all the materials have been placed in the mixer, shall be sufficient for thorough mixing to take place, but on no account be long enough for segregation of the constituent materials to occur. Unless uniformity tests in accordance with ASTM C94 are performed, a minimum mixing time of 1 ½ minutes for drum type mixers shall be allowed or such other time as the Owner shall direct. For batches larger than one cubic meter, mixing time shall be increased 15 seconds for each additional cubic meter or fraction thereof.
- 3) Cement when contained in bags, shall be used in such a way that each batch of concrete requires no part bags to be used. If the cement is stored in bulk, the measuring device shall be approved by the Owner before use.
- 4) The quantity of water for each batch shall be accurately measured and discharged to the mixer while the later is being charged.
- 5) All measuring devices shall be accurate to within $\pm 1\%$ and shall be proved to the satisfaction of the Owner before any concrete is mixed, and at such other times as may be directed by the Owner.
- 6) Concrete shall have the minimum water cement ratio as far as possible and shall be of such consistency that it can be readily worked into corners and angels of the formwork, and around the reinforcement without segregation of the materials or bleeding of the free water at the surface. On striking the formwork, it shall present a face which is uniform, free from honey combing, surface crazing or excessive dusting, and shall be to the approval of the Owner.
- 7) Ready-mixed concrete may be used if written consent is obtained from the Owner and shall be mixed and delivered in accordance with applicable requirements of ASTM C94, except that only trucks equipped with a rotary drum or agitator shall be used, and the concrete shall be deposited within the time limits specified in Clause 11.2.1.

- 8) The Bidder must have adequate plant to produce the required output, and leave satisfactory control of the mix proportions and the mixing of the concrete. The Owner shall have access to all parts of the plant during working hours.

11.2. CONVEYING CONCRETE

- 11.2.1** Concrete after being discharged from the mixer shall be transported as rapidly as possible, which in no case shall exceed 30 minutes, to its final position in its intended placement, by means that shall be approved by the Owner and which shall prevent adulteration, segregation, loss or contamination of the ingredients. The containers that convey the concrete shall at all times be kept clean and free from hardened or partially hardened concrete.
- 11.2.2** Chutes or other equipment for conveying concrete shall be of such size and design as to ensure a continuous flow of concrete at delivery end without separation of ingredients, and shall be thoroughly cleaned before each run. If use of pumping equipment to convey concrete is selected, aluminum pipe shall not be used.
- 11.2.3** Concrete shall not be allowed to drop freely more than 1.50 m. For drops exceeding 1.50 m a hopper and drop chute shall be used.

11.3. PREPARATIONS FOR PLACING CONCRETE

- 11.3.1** Before concrete is placed, clean the form and reinforcing of dirt, debris, and do all necessary trenching, damming and pumping so that all concrete can be placed in dry. Do not place any concrete until forms and reinforcing have been inspected and approved by the Owner.
- 11.3.2** Before fresh concrete is placed on or against hardened new concrete or old concrete, thoroughly clean surfaces of all laitance, soft or loose materials and deleterious substances. The surfaces shall be treated in accordance with the requirements of ACI 318, then be washed clean and thoroughly moistened.

Where concrete has dried out, it shall be saturated for at least 24 hours. Immediately before fresh concrete is deposited, horizontal surfaces shall be thoroughly covered with cement mortar of same mix as concrete to be poured. Fresh concrete shall then be placed before cement mortar has obtained its initial set.

- 11.3.3** All reinforcing and embedded steel shall be washed with clean fresh water using a high pressure jet to remove surface chloride contamination immediately prior to placement of the concrete. Washing shall be repeated if the steel is subsequently immersed in or contaminated by sea water.

11.4. PLACING OF CONCRETE

- 11.4.1** Concrete shall not be placed until all forms, embedded items and the surfaces upon which the concrete is to be placed have been cleaned up and approved by the Owner. Concrete shall be delivered to the forms by methods and equipment ensuring its required consistency without excessive loss of slump or segregation. Concrete shall be deposited in all cases as nearly as practicable directly in its final position and shall not be caused to flow in a manner to permit or cause segregation. Concrete shall not be moved to its final location by the use of vibrators.

- 11.4.2** The Owner shall be given 24 hours notice in order that he may check the work and no concrete shall be placed in any part of the structure until the Owner's approval has been given in writing.

If concreting is not started within 24 hours of approval being given, approval shall be obtained again from the Owner.

- 11.4.3** Place all concrete in a continuous pour over the area between construction joints in an uninterrupted operation in such manner as to form a monolithic structure, the component parts of which are securely bonded together. Do not deposit segregated concrete or concrete that has partially hardened or been contaminated by foreign materials, nor use retempered concrete. Time interval between placing of successive batches of concrete shall not be greater than 30 minutes.

- 11.4.4** Deposit all concrete in forms within 1.5 hours after introduction of mixing water to cement and aggregates unless otherwise required.

- 11.4.5** All concrete in vertical members, such as columns and walls, shall be in place not less than 2 hours before any concrete is placed in girders, beams or slabs to be supported thereon. Place concrete for beams, girders, brackets, column caps, and haunches monolithic with floor slabs.

11.4.6 Hot Weather Concreting

Hot weather concreting shall comply with ACI 305 and the following:

In hot weather suitable means shall be provided to ensure that the temperature of the concrete when it is placed in the work shall not exceed 32oC.

Where necessary the aggregate stockpiles shall be shielded from direct sunshine and/or sprayed with fresh water especially when evaporation rates are high. Due allowance shall be made for the extra water in the concrete mix design. Mixing water shall be kept cool with insulated pipes and open tanks, refrigerated or ice added if required.

Where necessary the surfaces (including reinforcement) against which concrete is to be placed shall be shielded from direct sunshine and sprayed with fresh water to prevent excessive absorption by the surfaces of water from the fresh concrete.

Concrete placing shall be completed as quickly and efficiently as possible, and exposed surfaces shall be immediately covered to reduce excessive evaporation of water from the concrete and conform to the applicable requirements of ACI 301.

The Bidder shall present for the Owner's approval his proposals for dealing with the following problems related to hot weather concreting:

- (i) Reduced workability
- (ii) Excessive plastic shrinkage
- (iii) Rapid strength gain but possible low final strength
- (iv) Rapid drying out of concrete
- (v) Concrete mixing to counteract hot weather
- (vi) Mixing water to counteract hot weather

11.4.7 Mass Concrete

For placing massive concrete, concrete shall be placed in layers approximately 450 mm thick, in accordance with ACI 301.

11.4.8 Underwater Concrete

Underwater concrete shall be considered as an unusual event. In general, the Bidder shall carry out full removal of excess water from excavations before starting pouring concrete and no concrete shall be placed in flowing water.

Under water concrete shall be placed in position by tremie tube specified in Clause 15 and the Bidder shall submit full details of the method proposed in advance to the Owner for approval before starting concreting.

11.4.9 Compaction of Concrete

Thoroughly compact all concrete work by means of mechanical vibrations, subject to the following minimum requirements:

- (a) Vibrator shall be electric or pneumatic power driven types and shall operate at frequency of 7,000 to 10,500 vibrations per minute for vibrating heads from 80 to 150 mm in diameter and from 8,000 to 13,500 vibrations per minute for vibrating heads of 100 mm or greater in diameter, when fully immersed.
- (b) Intensity of vibration shall be such as to visibly affect a mass of concrete of 250 mm slump over a radius of at least 450 mm.
- (c) The vibrators shall be inserted and withdrawn out of the concrete slowly. Application of vibrators shall be at points uniformly spaced and not farther apart than twice the radius over which the vibration is visibly effective. If requested by the Owner, the concrete shall be vibrated again.
- (d) Vibration shall not be applied directly or through the reinforcement to sections or layers of concrete which have hardened to the degree that the concrete ceases to be plastic under vibration.

- 11.4.10** Vibration shall continue until the concrete has been compacted to its maximum practicable density, free from pockets, closes snugly against all surfaces and is in perfect and complete contact with all reinforcement bars and other metal embedded in the concrete. The manufacturer of the vibrators and the total number of vibrator required shall be subject to the approval of the Owner.

12. SURFACE TREATMENT OF STRUCTURAL CONCRETE

12.1. GENERAL

“Surface Treatment” of structural concrete shall mean a finish which is an integral part of the structural concrete.

Surface treatment shall conform to requirements and uses specified in this Clause unless otherwise indicated.

12.2. EXPOSED SURFACE FINISH

12.2.1 Upon removal of forms, all fins and other projections shall be removed and offsets leveled. All voids, holes, honeycomb or other damaged surfaces shall be cleaned back to solid concrete, saturated with water and filled with cement mortar of same composition as that used in the concrete. The concrete shall be finished free from streaks, discolorations or other imperfections so as to produce an extremely smooth, dense and true finish of uniform color.

12.3. UNEXPOSED SURFACE FINISH

12.3.1 Upon removal of forms, all voids, holes, honeycomb or other damaged surfaces shall be cleaned back to solid concrete, saturated with water and filled with cement mortar of same composition as that used in the concrete.

12.4. ROUGH SLAB FINISH

12.4.1 Requirements

Top surface of structural concrete slabs and equipment foundations shall be level unless otherwise indicated. Screeds shall be used for all structural slabs and shall be securely and accurately set to elevations indicated. After concrete has been compacted by vibrating, it shall be screeded to required level.

After screeding, and when concrete has partially hardened so that it will retain impressions of a broom, it shall be broomed with a stiff-bristled broom and all laitance and scum removed. The brooming shall expose some of the aggregate and score the surface to provide mechanical bond for separate finish.

12.4.2 Uses

- Structural concrete floor slabs or bases where separate concrete floor finish, concrete floor-fill-finish, quarry tile, terrazzo or similar separate finishes are indicated or required.
- Top surface of equipments.
- Top of concrete duct runs.

12.5. MONOLITHIC FINISH

12.5.1 Requirements

Monolithic finish is one produced on structural concrete slabs which are poured all at one time with one concrete mix. Concrete for structural slabs with monolithic finish shall be of the dries consistency possible to work with a sawing motion of screed.

Surface shall first be screeded, and then be floated to a smooth, compact surface. Any water sheen shall be allowed to evaporate or removed prior to floating operation by covering surface with burlap or similar material. Spreading of cement or cement-sand mixture directly on top on concrete will not be permitted.

After floating, the surface of concrete shall be covered by floor handener and shall be steel troweled to a smooth, slatelike surface. Final troweling shall be done after concrete is so hard that no mortar accumulates on trowel and a ringing sound is produced as trowel is drawn over surface.

Finish surface shall not be marked off in areas or scored in any matter. A concrete floor surface hardener “Colorcron” or approved equivalent shall be applied in accordance with the manufacture’s instruction for permanently exposed floors.

12.5.2 Uses

- Structural concrete floor slabs where monolithic finish is indicated or required.
- Floor areas where resilient tile finishes (asphalt, rubber, vinyl etc.) are indicated or required.

12.6. FLOAT FINISH

12.6.1 Surface shall first be screeded, and then shall be machin finishing to a smooth, compact surface. After machin finishing, surface shall be floated after concrete has hardened sufficiently to retain the float finish and water sheen has disappear.

12.6.2 Uses

- Structural concrete roof slabs. Finish shall be free of all depressions or projection which would prevent proper application of roofing.
- Structural floor slabs only where specifically indicated or required.
- Top of concrete walls, curbs, piers, pads, pedestals, and outdoor equipment foundations.
- Manholes, sump pits, elevator pits etc. Roofs, if any, of these structures shall also have a float finish except that where such roofs are integral with other concrete work, the roofs shall have the same finish as the other concrete work.

12.7. STEEL TROWEL FINISH

The same requirements apply as for monolithic finish, except that consistency of concrete shall be as used for the portion of work for which steel troweled finish is specified.

Surface shall be screeded and then shall be floated to a smooth compact surface. After floating the surface of concrete shall be steel troweled to a smoth, slatelike surface.

This finish shall be used for stair treads.

Floor area where resilient tile finishes (asphalt, rubber, vinyl, etc) are indicated or required.

12.8. BROOM FINISH

The surface shall first be screeded, and then shall be provided machine finishing to a smooth, compact surface. Following the bull floating, the surface shall be broomed with a medium stiff bristled broom but only after the concrete has hardened sufficiently to retain the scoring. The strokes shall be square across the surface and shall be made as to produce regular scoring without tearing the surface or exposing any aggregate. The scoring shall run transverse to the direction of traffic.

This finish shall be used for sidewalks. It shall be used for other surfaces only if specifically indicated.

13. PROTECTION OF CONCRETE

13.1. PROTECTION OF CONCRETE

Fully protect all concrete from damage or injury during construction operations. Protect exposed external corners of concrete with wood strips securely fastened in place. Use tarpaulins or other coverings when necessary, or as requested, to completely enclose freshly finished concrete.

13.2. CLEANING OF SURFACES

After concrete is placed, all exposed surfaces which have been contaminated by concrete splashing, dripping, etc., caused by concrete work, shall be cleaned to restore to their original condition.

14. CURING

14.1. GENERAL

Use membrane curing unless otherwise indicated for all concrete work. Minimum curing periods as specified herein shall be observed before subjecting concrete to live loads, earth loads, traffic, etc. Sea water shall not be used for curing concrete but fresh water may be used as an alternative.

14.2. CURING PERIODS

The following curing periods shall apply when the temperature of surrounding air is above 15°C

- 14.2.1** Concrete floors and stair treads with
- Monolithic or steel trowel finish 10 days
- 14.2.2** Roof slabs 15 days pr up to built
up roofing is provided
- 14.2.3** All other concrete work 7 days
- 14.3. MEMBRANE CURING FOR GENERAL CONCRETE WORK**
- Provide membrane curing as follows for all concrete work except concrete floors or stair treads with monolithic or steel trowel finish:
- 14.3.1** Immediately after completions of required surface treatment of concrete, as specified in Clause 12, completely cover surfaces of concrete with membrane curing compound specified in Table S-340-1.
- 14.3.2** Where forms are left in place for the entire specified curing period, use of curing compound may be omitted provided that the warm weather forms will be wetted down with water if and when requested.
- 14.3.3** Where forms are stripped before completion of specified curing period, apply curing compound to such surfaces immediately after completion of specified surface treatment.
- 14.4. CURING COMPOUNDS CONTAINING VOLATILE THINNERS**
- Do not permit smoking, open flame or fires while material is being applied. Provide adequate ventilation. Equip workman with respirators if required.
- 14.5. MEMBRANE CURING FOR CONCRETE FLOOR AND TREAD FINISHES**
- Provide membrane curing as follows:
- 14.5.1** Cure, seal and dustproof surfaces of concrete floors or stair treads with monolithic or steel trowel finish immediately after finishing by applying one coat of membrane curing compound, specified in Table S-340-1.

14.5.2 Application

Prior to initial use of any specified compounds, arrange with manufacturer for manufacturer's technical representative to visit the site and give personal instructions on correct use of material.

14.5.3 Membrane Thickness

Apply at rate of approximately 8 square meters per liter, unless otherwise instructed by manufacturer.

14.5.4 Cure for not less than periods specified before applying any other surfacing or before opening to traffic.

15. TREMIE CONCRETE

15.1. CONCRETE

Concrete shall conform to the requirements specified for Grade 30 concrete unless otherwise indicated, except 10 % additional cement shall be used in the mix, and slump limit shall be 140mm.

15.2. TREMIE EQUIPMENT

Pipe or tube having minimum diameter of 250 mm constructed in sections with flanged coupling fitted with gaskets, and with hopper at top.

Support tremie so as to permit free movement of discharge and over entire top surface of work and so as to permit tremie to be rapidly lowered, when necessary to choke off or retard concrete discharge.

15.3. PLACING

15.3.1 Deposit concrete underwater by carefully placing concrete in its final position by means of tremie. Do not disturb concrete after it is deposited.

15.3.2 Use special care to maintain still water at point to deposit. Do not place tremie concrete in running water.

15.3.3 Carefully regulate consistency of concrete and use special care to prevent segregation of materials.

15.3.4 Place concrete in continuous horizontal layers, with thickness of each layer generally not exceeding 300 mm. Carefully regulate method of depositing concrete so as to produce approximately horizontal surfaces on each layer.

15.3.5 Have tremie hopper completely full of concrete during entire placing operation in order to maintain sufficient head. Keep flow continuous, and in no case interrupt until placing of concrete is completed.

16. CONCRETE FILL FOR EQUIPMENT BASES

Fill hollow based of equipment with concrete unless otherwise indicated, after bedplates have been grouted. Provide concrete fill conforming to requirements for class of concrete specified for the work, except maximum size coarse aggregate shall be 19 mm.

17. CAST-IN-PLACE ELECTRICAL DUCT RUNS

Conform to the pertinent requirements of Technical Specification for Electrical Works and use concrete conforming to requirements of class of concrete specified for the work, with maximum size of coarse aggregate 19 mm.

18. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality assurance /quality control shall conform to the requirements of Part 4, Subsection 2.7.

(Last page of S-340)